Wazuh SIEM Case Study Submission

Submitted To: CyberXchange security operations center
Submitted by: MENTIMI SANDEEP BABU
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1. Objective

The goal of this case study was to:

· Install Wazuh SIEM (Manager + Dashboard) on a local virtual machine.
· Deploy a Wazuh agent on a separate virtual machine and connect it to the Manager.
· Perform an Nmap scan on the agent machine to simulate a malicious reconnaissance activity.
· Verify that the SIEM successfully detects the scan and generates a security alert in the dashboard.

2. Lab Environment Setup

I used two virtual machines running on the same local network (Bridged mode) to complete this task:

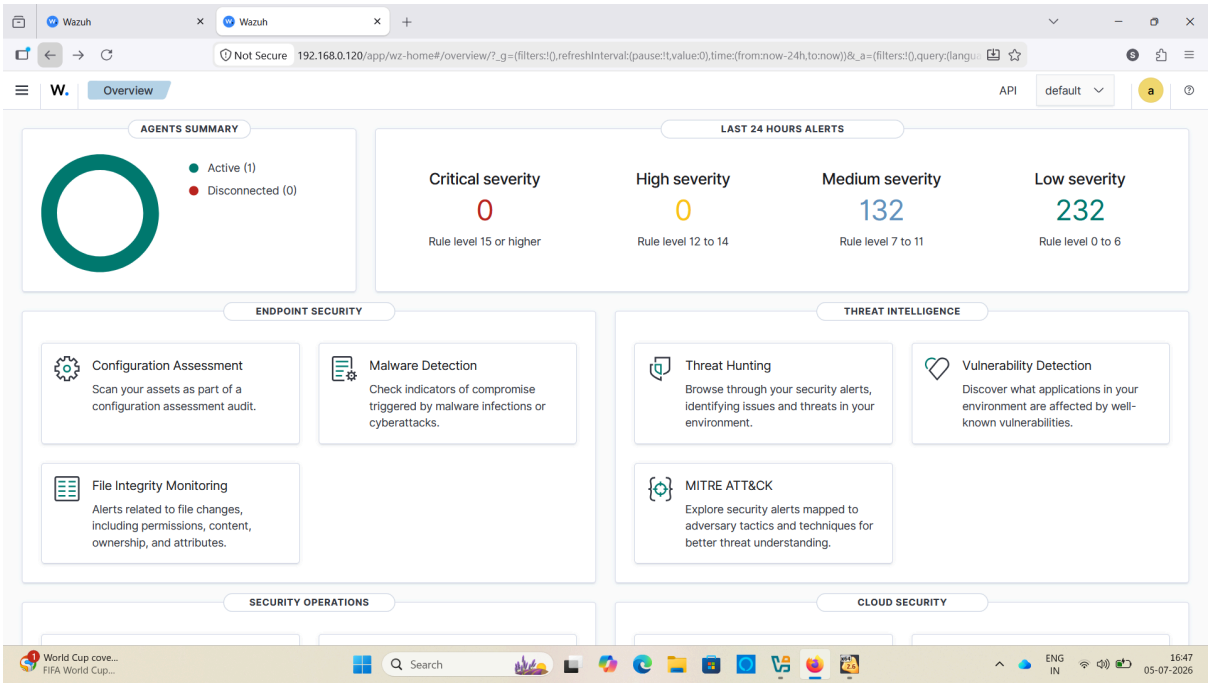
Component Operating System IP Address
Wazuh Manager (Server) Ubuntu 24.04 [Enter Manager IP, e.g., 192.168.0.120]
Wazuh Agent (Endpoint) Ubuntu 24.04 / Kali Linux [Enter Agent IP, e.g., 192.168.0.131]

3. Step-by-Step Implementation

Step 1: Installing the Wazuh Server (Manager VM)

· What I did:
  I installed the Wazuh central components (Indexer, Manager, and Dashboard) on a single machine using the official all-in-one installer.
· How I did it:
  On the Manager VM, I executed the following commands:
  ```bash
  curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
  sudo bash ./wazuh-install.sh -a
  ```

After installation, the script generated an admin password and provided the dashboard URL. I accessed the dashboard via https://[192.168.0.120]



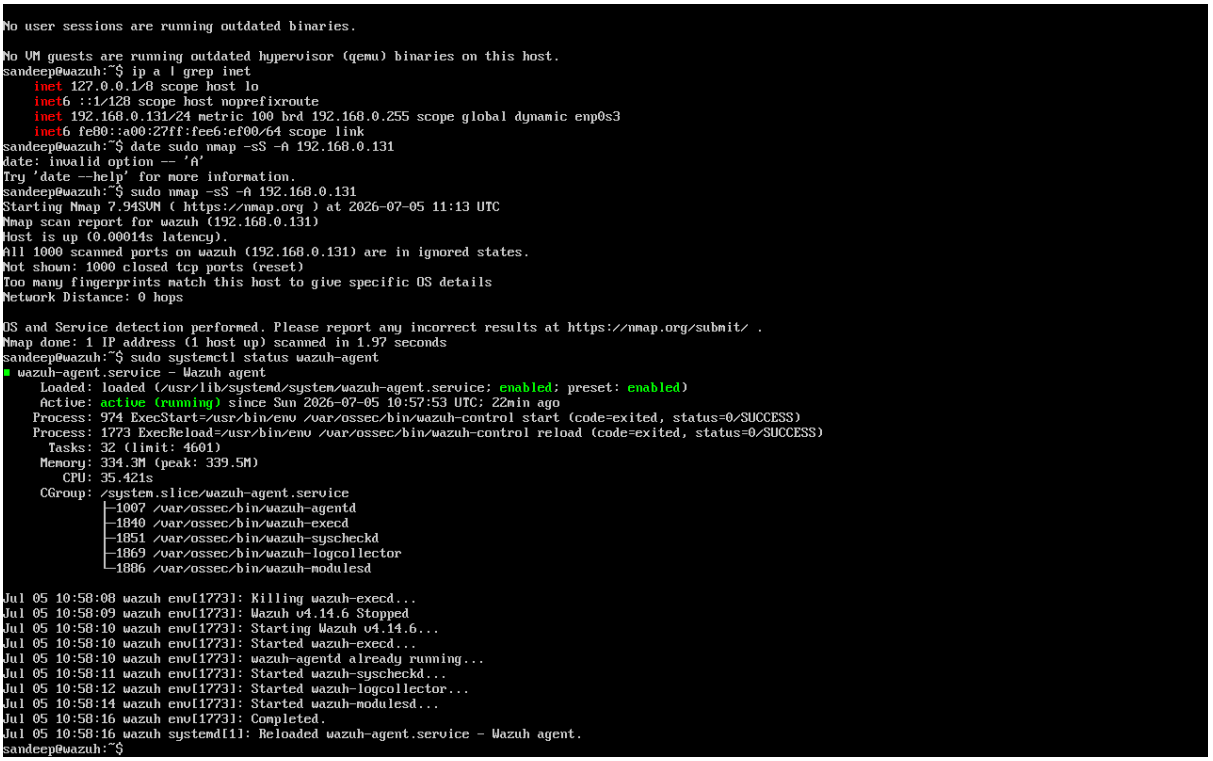
Step 2: Installing the Wazuh Agent on the Separate VM

· What I did:
I installed the Wazuh agent on the second virtual machine to act as the monitored endpoint.
· How I did it:
On the Agent VM, I downloaded the agent package and installed it while pointing it to the Manager's IP address:

```bash
wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.12.0-1_amd64.deb
sudo WAZUH_MANAGER='[192.168.0.120]' WAZUH_AGENT_NAME='my-agent' dpkg -i ./wazuh-agent_4.12.0-1_amd64.deb
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
```

Step 3: Verifying Agent Connectivity in the Dashboard

· What I did:
I checked whether the agent had successfully established a connection with the Wazuh Manager.
· How I did it:
I logged into the Wazuh Dashboard and navigated to Management → Agents. I verified that my agent appeared with a green Active status.



Step 4: Performing the Nmap Scan to Trigger an Alert

· What I did:
To simulate a malicious port scan, I ran an Nmap scan against the Agent VM itself.
· How I did it:
First, I installed Nmap on the Agent VM using sudo apt install nmap -y. Then, I ran a stealth SYN scan combined with service detection:

```bash
date
sudo nmap -sS -A localhost
```

(I included the date command right before the scan to clearly capture the timestamp in the terminal.)

```
    Process: 974 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Process: 1773 ExecReload=/usr/bin/env /var/ossec/bin/wazuh-control reload (code=exited, status=0/SUCCESS)
      Tasks: 32 (limit: 4601)
     Memory: 334.3M (peak: 339.5M)
        CPU: 35.421s
     CGroup: /system.slice/wazuh-agent.service
             ├─1007 /var/ossec/bin/wazuh-agentd
             ├─1840 /var/ossec/bin/wazuh-execd
             ├─1851 /var/ossec/bin/wazuh-syscheckd
             ├─1869 /var/ossec/bin/wazuh-logcollector
             └─1886 /var/ossec/bin/wazuh-modulesd

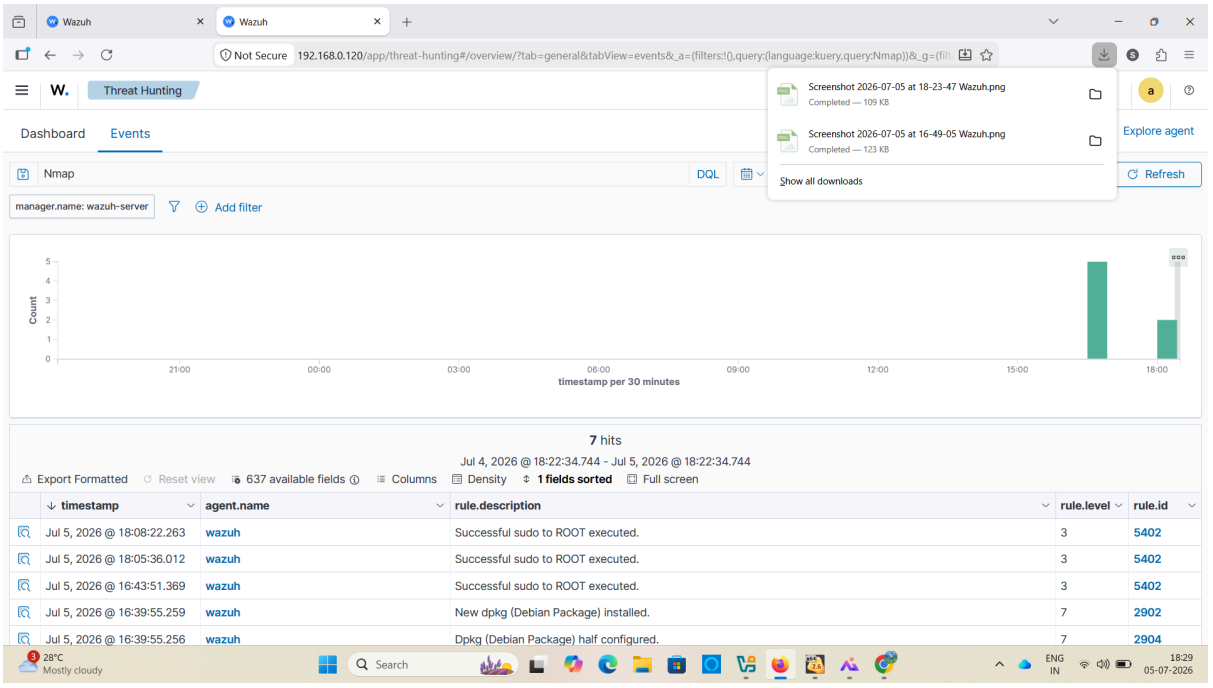
Jul 05 10:58:08 wazuh env[1773]: Killing wazuh-execd...
Jul 05 10:58:09 wazuh env[1773]: Wazuh v4.14.6 Stopped
Jul 05 10:58:10 wazuh env[1773]: Starting Wazuh v4.14.6...
Jul 05 10:58:10 wazuh env[1773]: Started wazuh-execd...
Jul 05 10:58:10 wazuh env[1773]: wazuh-agentd already running...
Jul 05 10:58:11 wazuh env[1773]: Started wazuh-syscheckd...
Jul 05 10:58:12 wazuh env[1773]: Started wazuh-logcollector...
Jul 05 10:58:14 wazuh env[1773]: Started wazuh-modulesd...
Jul 05 10:58:16 wazuh env[1773]: Completed.
Jul 05 10:58:16 wazuh systemd[1]: Reloaded wazuh-agent.service - Wazuh agent.
sandeep@wazuh:~$ [ 5541.285320] e1000 0000:00:03.0 enp0s3: Reset adapter
^C
sandeep@wazuh:~$ sudo nmap -sS -A 192.168.0.131
[sudo] password for sandeep:
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-05 12:35 UTC
Nmap scan report for wazuh (192.168.0.131)
Host is up (0.000056s latency).
All 1000 scanned ports on wazuh (192.168.0.131) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.53 seconds
sandeep@wazuh:~$ date
Sun Jul  5 12:38:16 PM UTC 2026
sandeep@wazuh:~$ sudo nmap -sS -A 192.168.0.131
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Nmap done: 1 IP address (1 host up) scanned in 2.40 seconds
sandeep@wazuh:~$
```

Step 5: Viewing the Generated Alert in the SIEM Dashboard

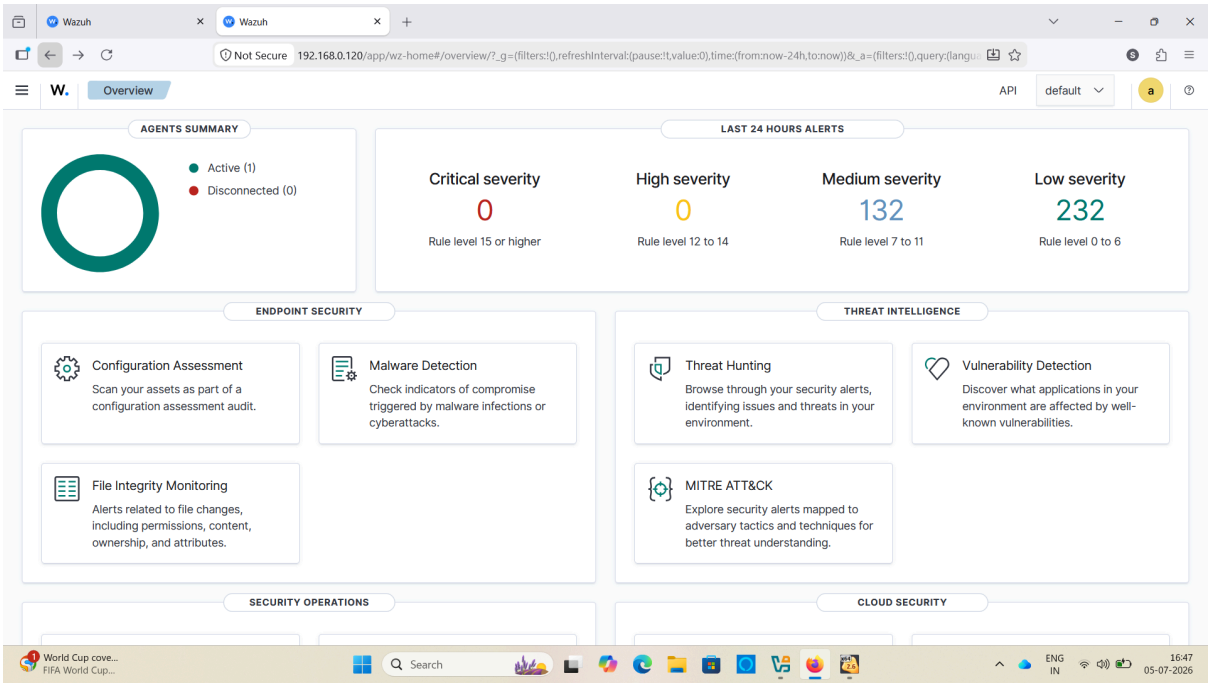
· What I did:
I confirmed that Wazuh detected the Nmap activity and generated a corresponding security alert.
· How I did it:
I went to the Threat Hunting → Events module in the dashboard. I applied a filter for my agent name (agent.name:wazuh) and searched for nmap. The event clearly showed the source IP, destination IP, and the exact timestamp of the scan.



4. Conclusion

The case study was executed successfully. The Wazuh environment was fully deployed, the agent was enrolled without errors, and the Nmap scan was instantly detected by the SIEM. The alert appeared in the dashboard with accurate metadata, proving that the detection rules are functioning properly. This exercise demonstrates a complete Security Operations workflow from endpoint monitoring to threat detection.

screenshots:

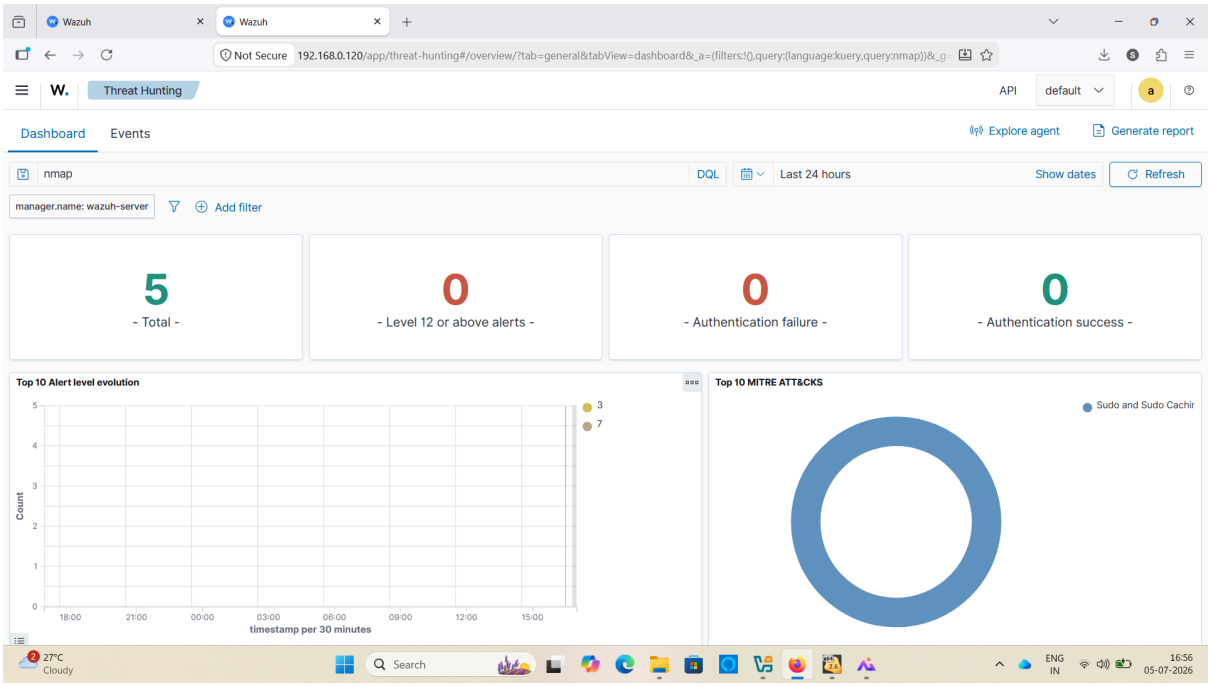



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```


Wazuh
Wazuh
Not Secure
192.168.0.120/app/threat-hunting#/overview/?tab=general&tabView=dashboard&_a=(filters:!(),query:(language:kuery,query:nmap))&_g
Threat Hunting
API
default
a
Dashboard
Events
Explore agent
Generate report
nmap
DQL
Last 24 hours
Show dates
Refresh
manager.name: wazuh-server
Add filter
5
- Total -
0
- Level 12 or above alerts -
0
- Authentication failure -
0
- Authentication success -
Top 10 Alert level evolution
Count
timestamp per 30 minutes
3
7
Top 10 MITRE ATT&CKS
Sudo and Sudo Cachir
27°C
Cloudy
Search
ENG
IN
16:56
05-07-2026